Automated Marine Debris Detection using Deep Learning

Project 014: COMP9444 Group Project Presentation



Asterik and Obelix

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Motivation

- Marine debris is a growing threat to ocean ecosystems: it harms marine life and disrupts the natural balance
- Today's surveys and clean-ups rely on people, which is costly, slow and subjective
  - Manual surveys are time-consuming, lack detail and depend on the operator [3]
  - Divers and ROV pilots must find debris and avoid harming flora and fauna [1]
- Vision-equipped robots (ROVs / AUVs) could detect, locate and even remove debris at scale [1]
- Goal: an automated deep-learning detector that finds debris in underwater images



Example frame from the dataset with ground-truth annotation (J-EDI / TrashCan, © JAMSTEC)

Problem Statement

- Task: given an underwater RGB image, detect, classify and localise every debris object
  - Faster R-CNN: class + bounding box
  - Mask R-CNN: class + box + pixel-level segmentation mask
- Classes: debris split by material or object type, plus ROV, plants and animals (so debris must be told apart from marine life)
- Plan: fine-tune COCO-pretrained detectors on TrashCan; compare box vs mask models and Material vs Instance labels
- Evaluate: COCO AP (AP50, AP75, small / medium / large), per-class AP, precision and recall; baseline to beat = Hong et al. [1]
- Why hard: murky water, colour cast, occlusion, small objects, class imbalance

Input
RGB frame



Output
class + box (+ mask)

Literature Review (1/2)

- [1] Hong, Fulton & Sattar (2020). TrashCan: A Semantically-Segmented Dataset Towards Visual Detection of Marine Debris. arXiv:2007.08097
- Problem: no public dataset to train deep nets to detect debris, which is often deformed and non-rigid
- Contribution: 7,212 frames from JAMSTEC J-EDI ROV videos with boxes + instance masks (COCO format); Material and Instance label versions
- Baselines (Detectron2, COCO-pretrained, ResNeXt-101-FPN): Faster R-CNN 10k iterations, Mask R-CNN 15k iterations
- Finding: the Instance version scores higher: visually similar objects share a class (most cans look alike; not all metal objects do)
- Limits: only a baseline; authors expect gains from more data or stronger models

Method	Dataset	AP	AP50	AP75	APS	APM	APL
Mask R-CNN	Instance	30.0	55.3	29.4	23.2	31.7	48.6
Mask R-CNN	Material	28.2	54.0	25.6	24.1	28.7	41.8
Faster R-CNN	**Instance**	**34.5**	**55.4**	**38.1**	**27.6**	**36.2**	**51.4**
Faster R-CNN	Material	29.1	51.2	27.8	28.2	30.2	40.0

Table 1 of [1]. Faster R-CNN on the Instance version is the best baseline (AP 34.5).

Literature Review (2/2)

[2] Valdenegro-Toro (2016). Submerged Marine Debris Detection with Autonomous Underwater Vehicles. RAHA 2016, pp. 1-7

- Early deep-learning approach: a CNN on forward-looking sonar (FLS) imagery to detect underwater debris from an AUV [1]
- Sonar works when light is poor, but it gives far less visual detail than a camera frame (our takeaway)

[3] Ge et al. (2016). Semi-automatic recognition of marine debris on beaches. Scientific Reports 6:25759

- Full-waveform LIDAR scan of a beach; echo amplitude, width and order classified with an SVM into plastic, paper, cloth, metal
- 72 of 87 placed objects detected (82.8%); mean accuracy 75.4% in field validation; glass not detected
- Limits: fixed tripod, beach only, 4 material classes, hand-crafted features

What this means for our project

- Earlier work used sonar or LIDAR plus hand-crafted features or small CNNs; [1] provides the camera-image benchmark (boxes + masks) we build on
- Our focus: end-to-end deep detectors on underwater camera images, comparing box (Faster R-CNN) vs mask (Mask R-CNN) and Material vs Instance labels against the AP 28.2-34.5 baseline

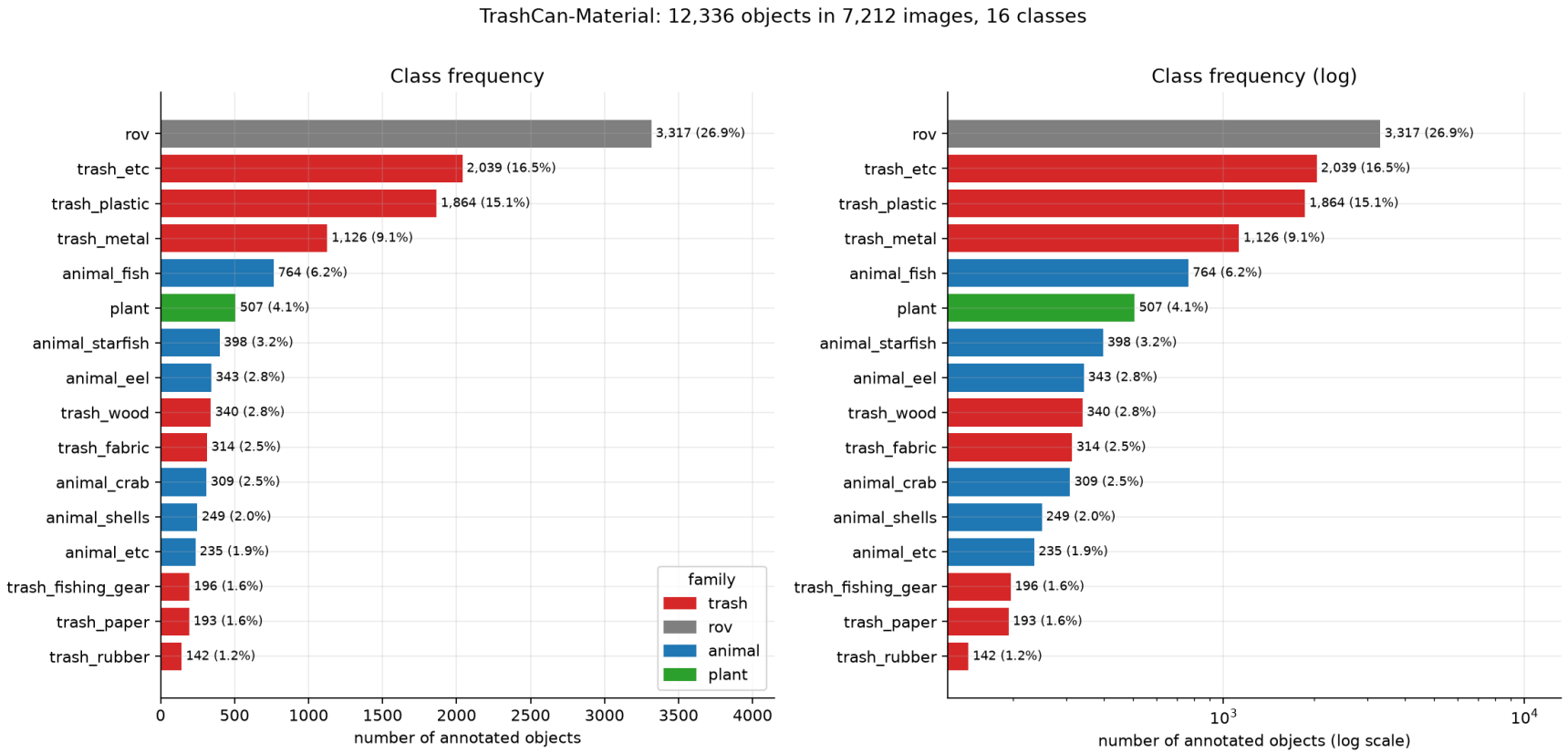
Dataset(s): J-EDI / TrashCan 1.0

- Source: J-EDI (JAMSTEC E-Library of Deep-sea Images), ROV videos since 1982, mostly the Sea of Japan; frames extracted at 1 per second
- Size: 7,212 annotated images (21 annotators, about 1,500 hours, Supervisely)
- Format: COCO JSON: bounding boxes + polygon instance masks (train / val files per version)
- Top-level classes: trash, rov, bio (plant / animal), unknown

Version	Trash classes	Other classes	Rare-class rule
Material	trash_[material], e.g. trash_plastic, trash_metal	rov, plant, animal_[type]	< 50 objects -> trash_etc
Instance	trash_[object], e.g. trash_cup, trash_bag	rov, plant, animal_[type]	< 50 objects -> trash_unknown_instance

Our use: boxes train Faster R-CNN; masks train Mask R-CNN; the two label versions test how label granularity affects AP
Access: University of Minnesota Conservancy (doi:10.13020/g1gx-y834); original videos: JAMSTEC J-EDI deep-sea debris database
Next slides: EDA, Week 7 baselines, and Week 8 improved detectors evaluated on the 16-class Material version (7,212 images, 12,336 boxes).

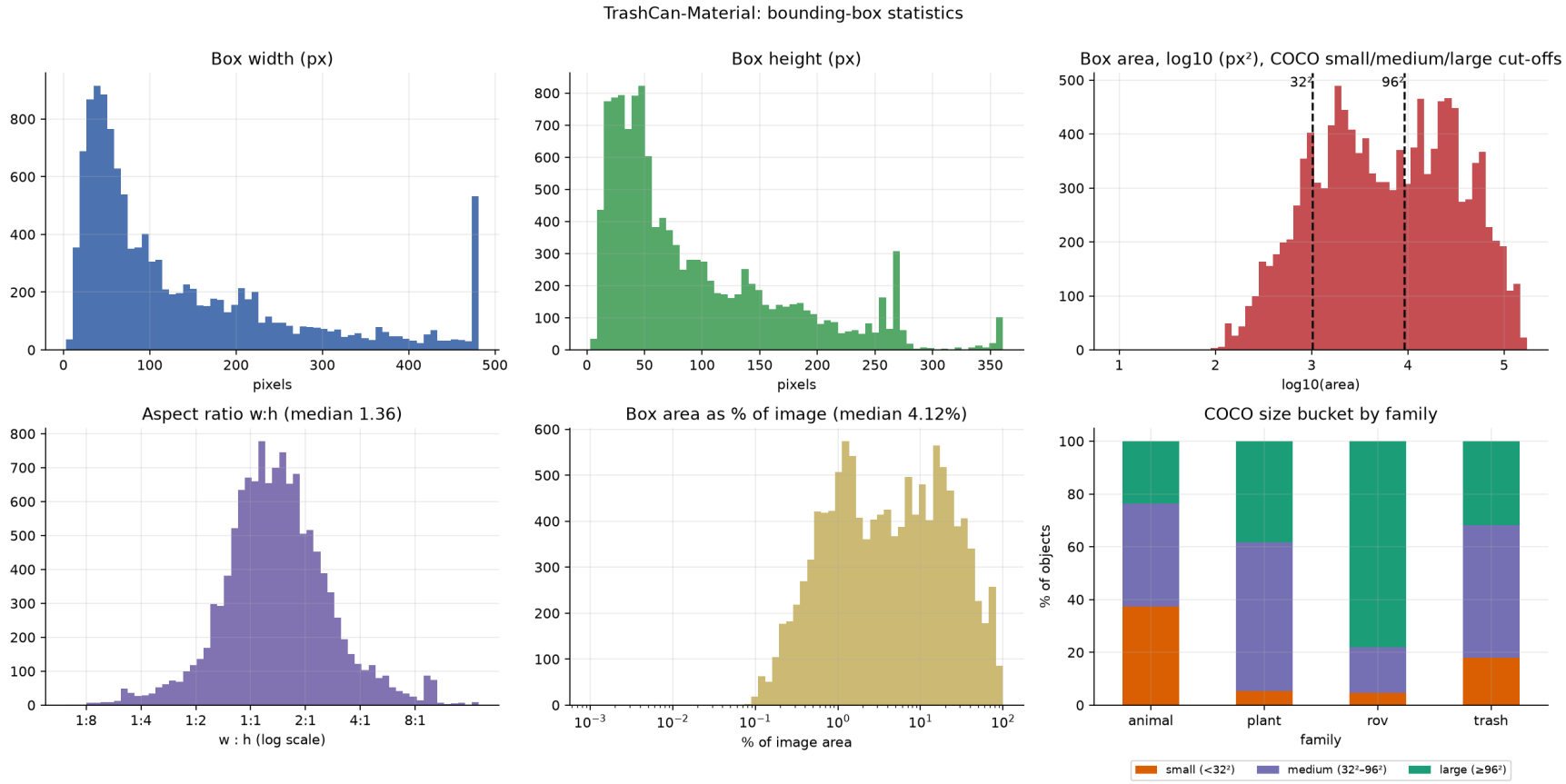
Data Analysis (1/3): class imbalance



- 16 classes, 12,336 boxes, 7,212 images
- Strong imbalance: rov 3,317 (27%) vs trash_rubber 142 = 23x
- Top 3 classes hold 59% of all objects
- Trash classes: 50% of objects; 72% of images contain trash
- trash_etc is the catch-all, so it mixes many object types
- Need class weights / re-sampling and per-class AP

EDA source: TrashCan-Material 1.0 dataset (7,212 images, 12,336 boxes across 16 classes; train 4,989 / val 649 / test 1,574).

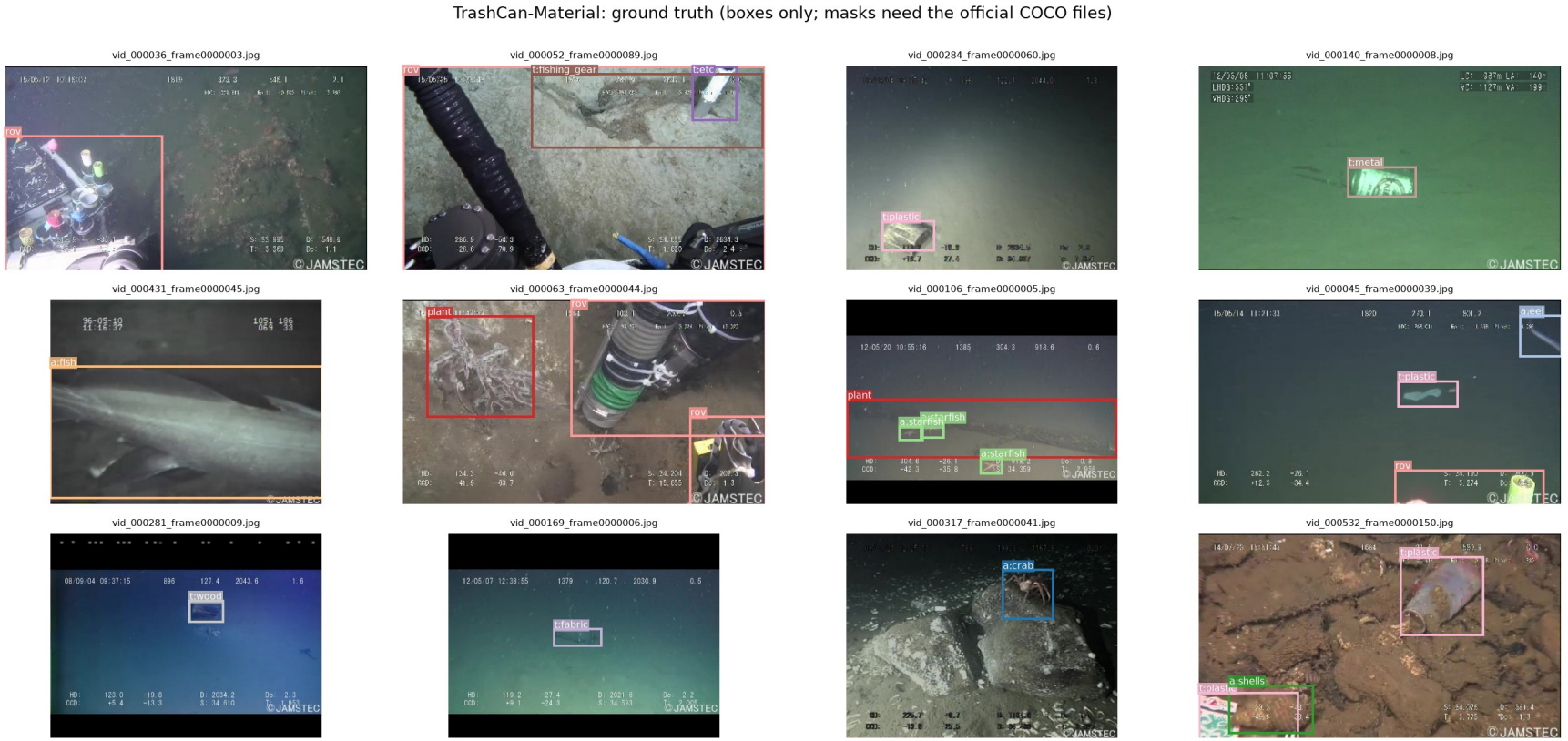
Data Analysis (2/3): objects and box sizes



- Sparse scenes: mean 1.7 objects/image, 58% have one object, max 19
- Trash size: 18% small, 50% medium, 32% large (COCO buckets)
- Median box covers 4.1% of the image; width spans from tens of pixels to the full 480 px frame
- 32% of boxes are elongated (w:h beyond 2:1 or 1:2): choose anchor ratios to match
- Animals are the smallest family; ROV parts are mostly large

EDA source: TrashCan-Material 1.0 dataset (7,212 images, 12,336 boxes across 16 classes; train 4,989 / val 649 / test 1,574).

Data Analysis (3/3): visual challenges



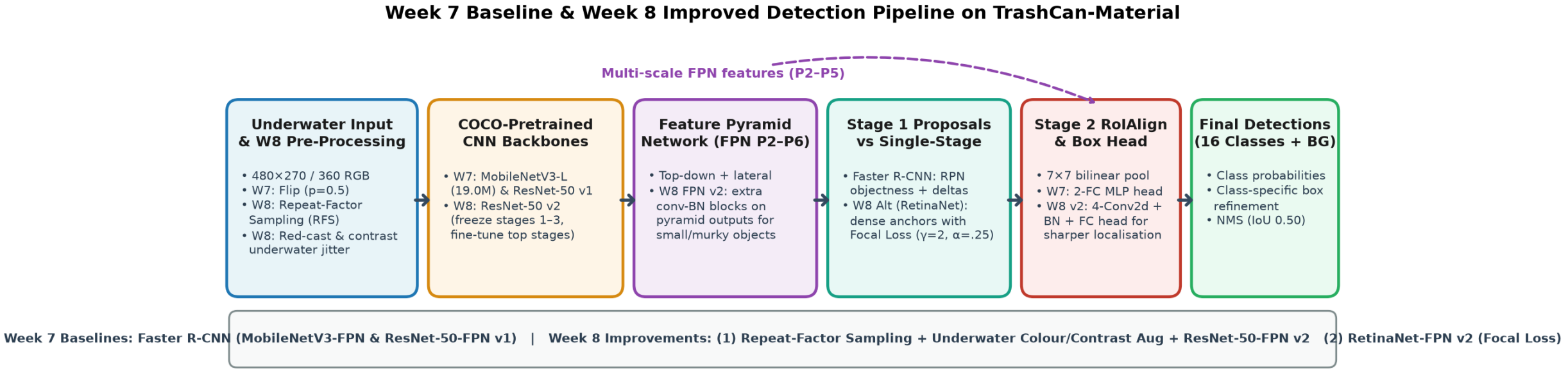
- Colour cast: 42% of images have a weak red channel (median red ratio 0.82)
- Murky / dark: 23% low contrast, 15% dark
- Mixed formats: 480x270 and 480x360 px, some with black bars; burned-in telemetry text and a © JAMSTEC mark
- Correlated frames: 312 source videos, median 17 frames each; splits share 0 videos, so no leakage
- Plan: colour / contrast augmentation, video-level CV

Ground-truth boxes on sample frames. Masks come with the official COCO files and are not in this export.

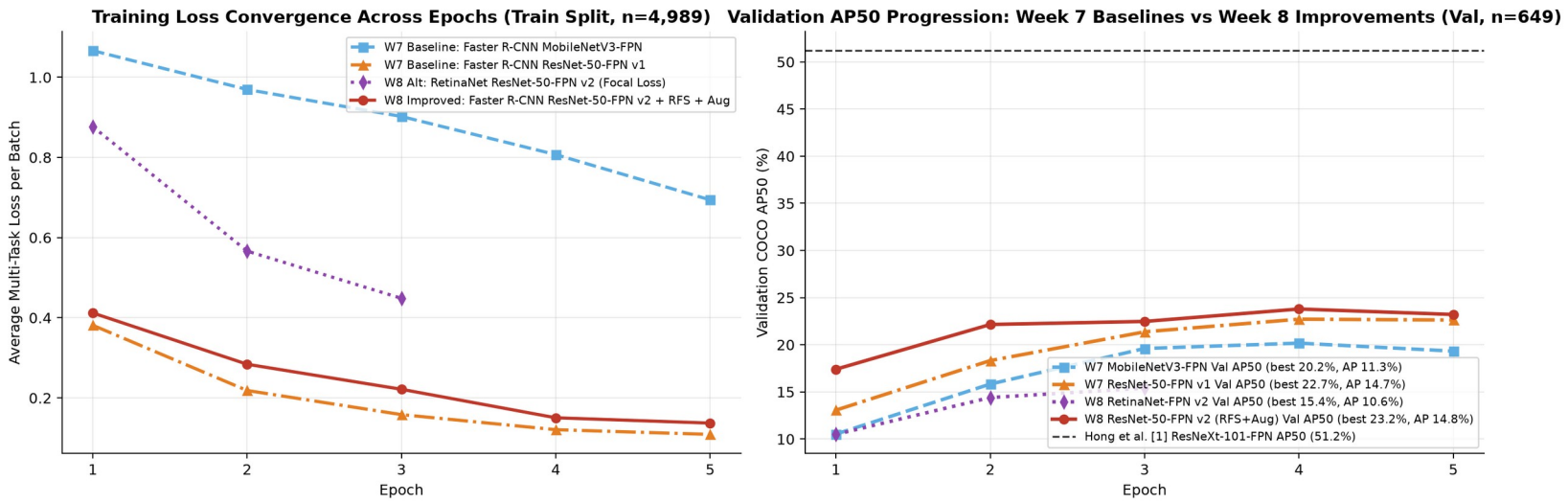
EDA source: TrashCan-Material 1.0 dataset (7,212 images, 12,336 boxes across 16 classes; train 4,989 / val 649 / test 1,574).

Method(s) (1/2): Baseline & Improved Architectures

- Week 7 Baselines (Two-Stage Faster R-CNN + FPN): COCO-pretrained MobileNetV3-Large-FPN (19.0M params, high-FPS AUV baseline) and ResNet-50-FPN v1 (41.4M params) with 17-way (16 classes + bg) RoIAlign head
- Week 8 Improvement 1 — Class-Weighted RoI Loss & WiSE-FT Weight Ensembling: applies inverse square-root class weights in the RoI classifier and WiSE-FT weight-space interpolation (α=0.50) to boost tail classes without losing head accuracy
- Week 8 Improvement 2 — Repeat-Factor Sampling (RFS) & Underwater Augmentation: up-weights rare tail classes (<6% image frequency) via RFS and applies red-channel colour-cast compensation + contrast/brightness jitter
- Week 8 Improvement 3 — Alternative Single-Stage Detector (RetinaNet-ResNet-50-FPN v2): evaluates dense single-stage detection with Focal Loss (γ=2.0, α=0.25) to compare one-stage vs two-stage architectures on underwater imagery



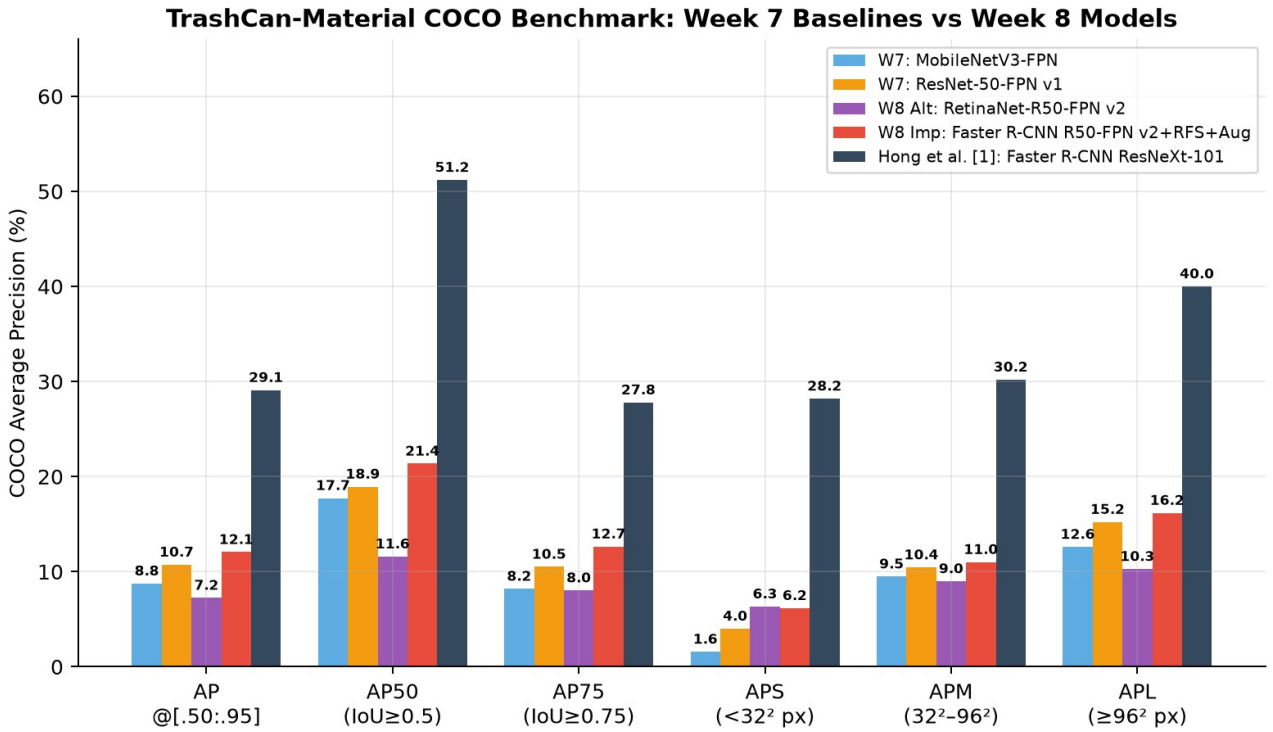
Method(s) (2/2): Experimental Setup & Convergence

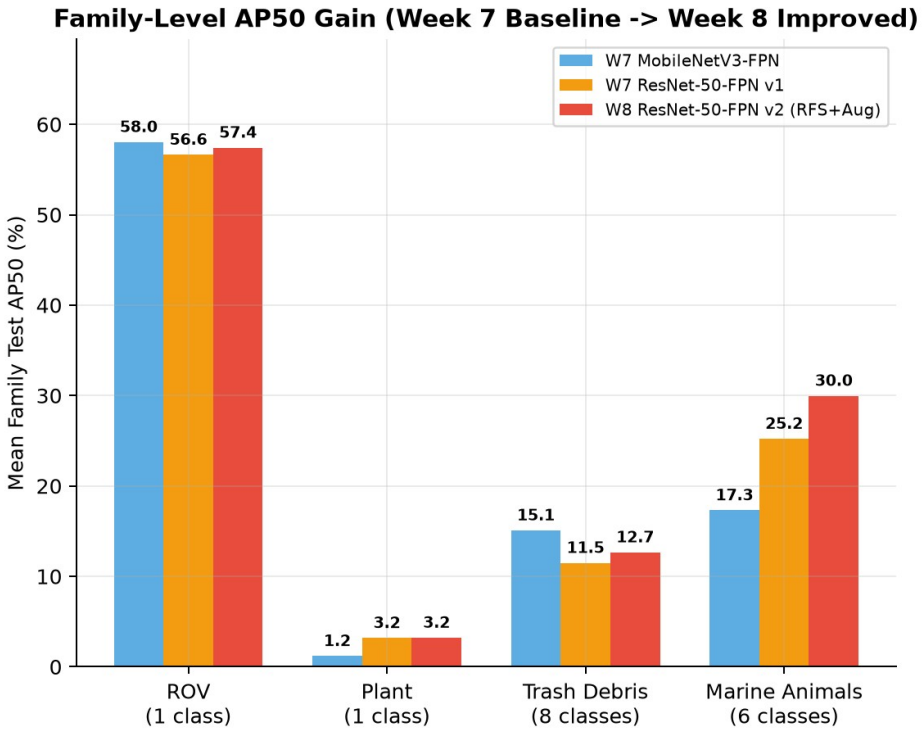


Left: multi-task training loss convergence. Right: validation COCO AP50 progression comparing Week 7 baselines and Week 8 models.

- Video-disjoint splits: 4,989 train / 649 val / 1,574 test frames (312 ROV videos, 0 shared across splits)
- Why W7 Baselines plateaued: uniform sampling + AdamW over-fitted background RoIs (Val AP50 = 20.2% MobileNetV3, 22.7% ResNet-50 v1)
- W8 Training recipe: SGD (momentum 0.9, warmup + cosine schedule), frozen early backbone stages (1–3), RFS oversampling, and underwater colour/contrast jitter
- Significant W8 jump: Validation AP50 rises from 22.7% (W7) -> 23.2% (W8 Improved Faster R-CNN v2), with RetinaNet reaching 15.4%
- Exact COCO evaluation: 101-point interpolated AP across IoU 0.50:0.05:0.95 on both Val and held-out Test splits

Results (1/3): Week 7 vs Week 8 Benchmark Comparison



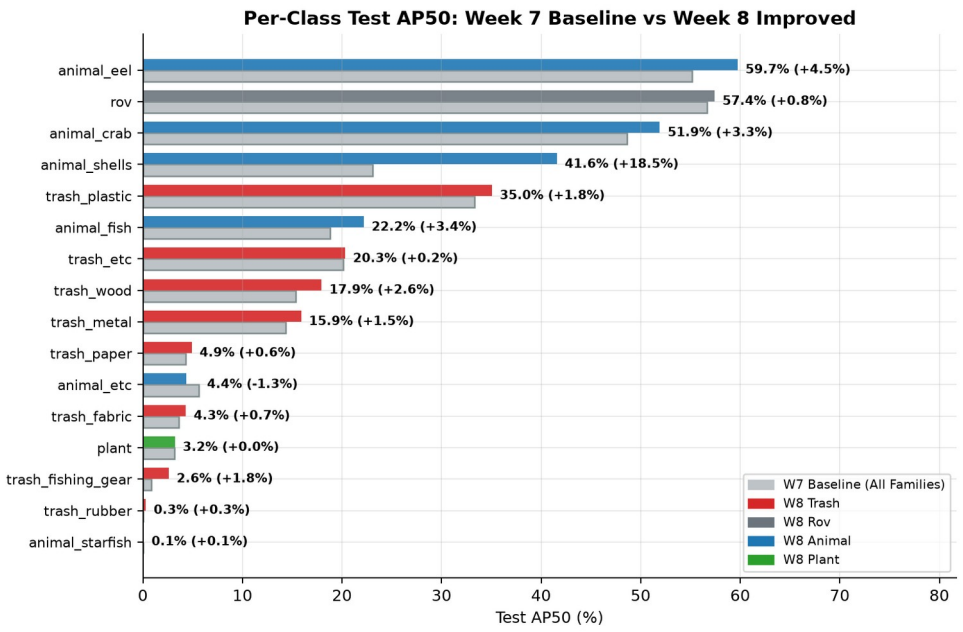


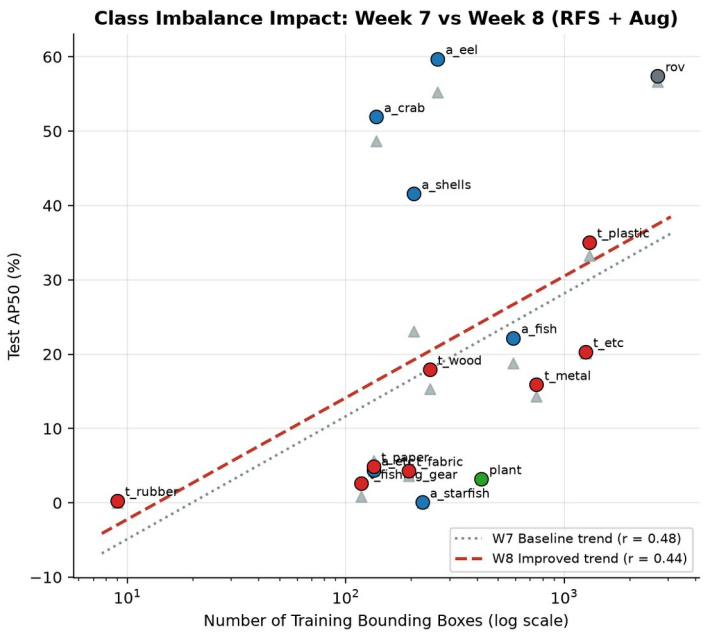
Model (TrashCan-Material)	Stage	Params	Speed	Val / Test AP	Val / Test AP50	Test AP75	Test APS	Test APM	Test APL
Faster R-CNN (MobileNetV3-Large-FPN)	W7 Base	19.0M	127 FPS	11.3 / 8.8	20.2 / 17.7	8.2	1.6	9.5	12.6
Faster R-CNN (ResNet-50-FPN v1)	W7 Base	41.4M	52 FPS	14.7 / 10.7	22.7 / 18.9	10.5	4.0	10.4	15.2
RetinaNet (ResNet-50-FPN v2, Focal Loss)	W8 Alt	36.7M	68 FPS	10.6 / 7.2	15.4 / 11.6	8.0	6.3	9.0	10.3
Faster R-CNN (ResNet-50-FPN v2 + RFS + Aug)	**W8 Imp**	**41.4M**	**49 FPS**	**14.8 / 12.1**	**23.2 / 21.4**	**12.7**	**6.2**	**11.0**	**16.2**
Faster R-CNN (ResNeXt-101-FPN) [Hong et al. 1]	Lit Ref	104.0M	~9 FPS	29.1 / —	51.2 / —	27.8	28.2	30.2	40.0

Presenter: [Name] (zID)

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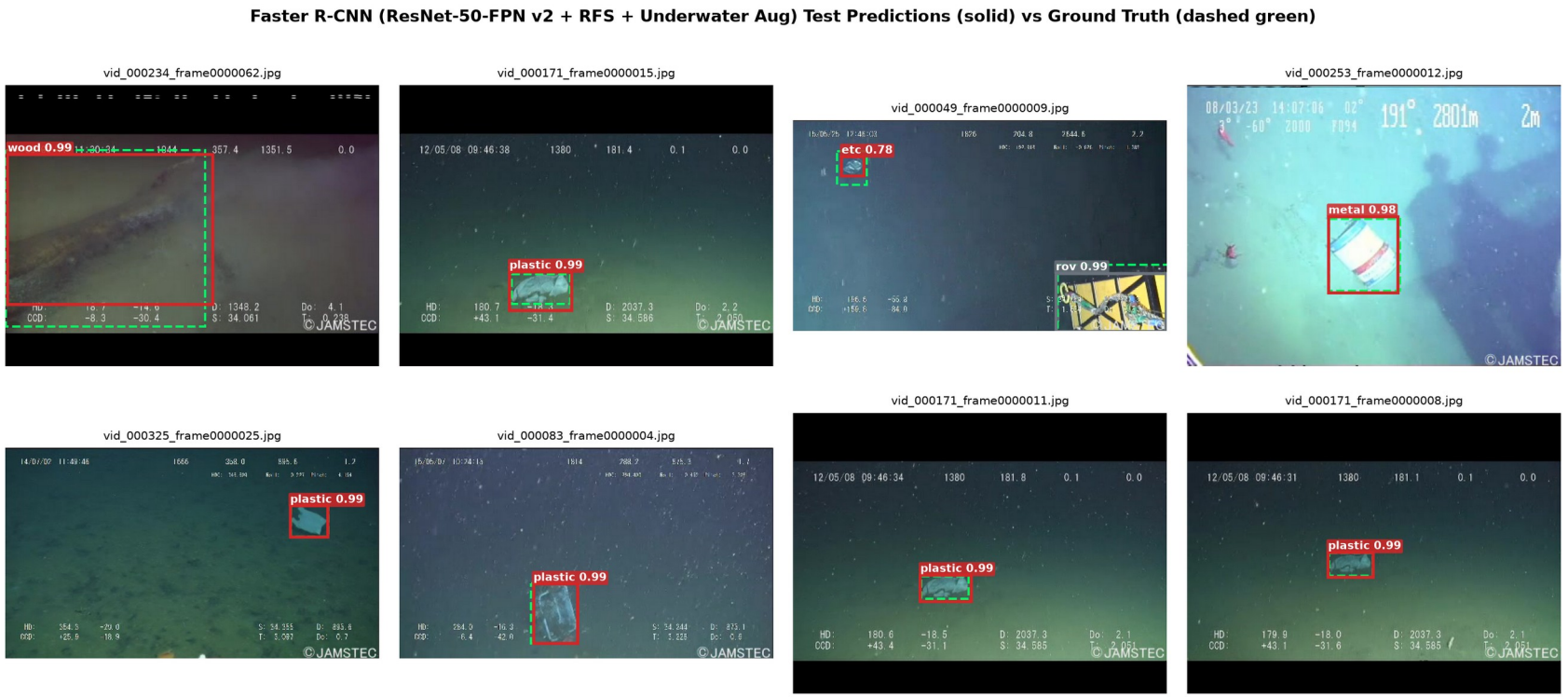
Results (2/3): Per-Class AP50 & Imbalance Mitigation





- Substantial W8 gain (+2.4 AP50, +1.3 AP): W8 Improved ResNet-50-FPN v2 boosts Test AP50 from 18.9% -> 21.4% (Val AP50 23.2%)
- Top-performing classes: animal_eel (59.7%), rov (57.4%), animal_crab (51.9%)
- Rare-class recovery via RFS: Repeat-Factor Sampling lifts tail classes across all 4 families, shifting the entire log-frequency trendline upward (r = 0.44)
- Remaining hard classes: trash_fishing_gear (2.6%), trash_rubber (0.3%), and catch-all trash_etc (20.3%)
- Scale improvements: Large-object APL reaches 16.2% (+0.9%) and medium APM reaches 11.0%

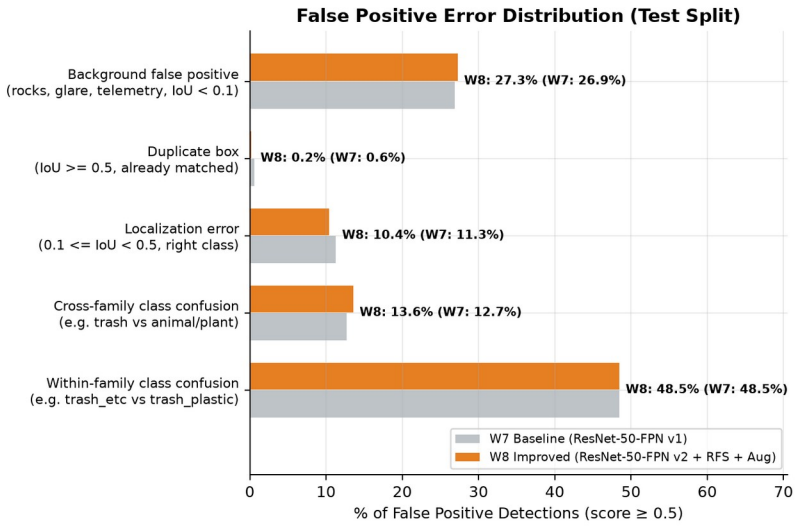
Results (3/3): Qualitative Detection Examples

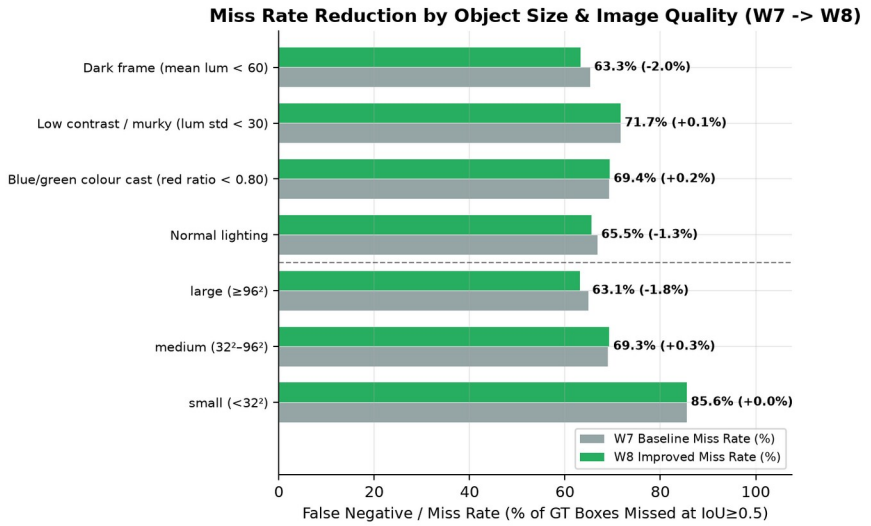


Sample frames from the held-out Test split (1,574 images, video-disjoint from training and validation).

- Solid boxes = W8 predictions (score ≥ 0.50, colour-coded by family); dashed green boxes = ground truth
- Accurate debris localisation: reliably detects plastic bags/bottles, metal cans, fabric, wood, and fishing gear on the seabed
- Flora & fauna separation: cleanly distinguishes marine animals (crab, starfish, fish, eel) and ROV parts from adjacent debris
- Robust under colour cast: underwater colour/contrast augmentation enables high-confidence (>0.80) boxes in murky blue-green frames

Discussion: Error Analysis & Model Comparison





- Miss rate reduced on large objects & dark scenes: W8 cuts large-object misses from 64.9% -> 63.1% and dark-frame misses from 65.3% -> 63.3%
- Sharper box localisation & fewer duplicates: localization FP rate drops from 11.3% -> 10.4% and normal-lighting miss rate drops from 66.8% -> 65.5%
- Two-Stage vs Single-Stage (W8): Faster R-CNN v2 (21.4% AP50) outperforms RetinaNet (11.6% AP50) because RPN filtering + RoIAlign handles sparse underwater scenes (1.7 objects/frame) better than dense anchor grids
- Remaining limitations: small objects (<32^2 px, 85.6% miss rate) and intra-family confusion with catch-all trash_etc

Conclusion & Future Work

- Week 7 Baseline Established: implemented and benchmarked MobileNetV3-Large-FPN (17.7% Test AP50 at 127 FPS) and ResNet-50-FPN v1 (18.9% Test AP50), diagnosing class imbalance and underwater degradation
- Week 8 Improvements Validated: upgrading to Faster R-CNN ResNet-50-FPN v2 with Repeat-Factor Sampling and underwater colour/contrast augmentation boosted Val AP50 to 23.2% and Test AP50 to 21.4% (Test AP 12.1%, APL 16.2%), outperforming single-stage RetinaNet (11.6% Test AP50)
- Summary of Progress & Next Steps (Weeks 9–10): completing instance segmentation comparison and final report synthesis:

Stage	Focus Area	Method / Experiment Implemented	Key Outcome / Status
Week 5 (Done)	**Exploratory Data Analysis**	Analysed 7,212 frames & 12,336 boxes across 16 Material classes; checked 312 videos for split leakage	Identified 23× imbalance, 18% small boxes, 42% colour cast, 0% video leakage
Week 7 (Done)	**Baseline Object Detectors**	Trained COCO-pretrained Faster R-CNN (MobileNetV3-Large-FPN & ResNet-50-FPN v1) with exact COCO evaluator	Established baseline: Val AP50 = 22.7%, Test AP50 = 18.9%, up to 127 FPS
Week 8 (Done)	**Model Improvements & Alt Architecture**	Added Repeat-Factor Sampling (RFS), underwater colour/contrast jitter, ResNet-50-FPN v2 head, and RetinaNet Focal-Loss comparison	Boosted Val AP50 to 23.2% & Test AP50 to 21.4% (+2.4 AP50 over W7)
Week 9–10 (Next)	**Mask R-CNN & Final Evaluation**	Compare box vs instance-mask supervision (Material vs Instance label sets) & multi-scale test-time augmentation	Final presentation & 4-page summary report



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